Marking Scheme

2026 · Paper Paper 1 · 105 marks

General Rules
• M / A / PROOF mark types
• **f.t.**： M marks always follow through. A marks follow through ONLY when isFollowThrough = true.
• Deduction cap： A(1)=1 · A(2)=1 · B=0 (u, pp)
• No deduction in any step scoring 0.
• A correct answer merits all marks unless a method is specified.
• Different notation is NOT penalized. · Benefit of doubt
• All fractional answers must be simplified.

Section A(1) (35 marks)

1. Laws of Indices [B-]

a (3 marks)

M1 $(m^3n^{-2})^{-3} = m^{-9}n^6$

M1 $(m^{-4}n)^2 = m^{-8}n^2; \dfrac{m^{-9}n^6}{m^{-8}n^2} = m^{-1}n^4$

A1 $\dfrac{n^4}{m}$

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2. Change of Subject [B-]

a (3 marks)

M1 $2u + 7 = v(5 - u)$

M1 $2u + uv = 5v - 7 \Rightarrow u(v + 2) = 5v - 7$

A1 $u = \dfrac{5v - 7}{v + 2}$

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3. Factorization [B]

a (1 mark)

A1 $3x^2 + 5xy - 2y^2 = (3x - y)(x + 2y)$

b (2 marks)

M1 $= (3x - y)(x + 2y) + 3(3x - y)$ f.t.
1M for extracting the common factor 3x - y

A1 $= (3x - y)(x + 2y + 3)$ f.t.

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4. Inequalities [B]

a (3 marks)

M1 $x + 7 > 3x - 3 \Rightarrow 10 > 2x \Rightarrow x < 5$

M1 $2x + 9 \geq 3 \Rightarrow 2x \geq -6 \Rightarrow x \geq -3$

A1 $-3 \leq x < 5$

b (1 mark)

A1 $1, 2, 3, 4; there are 4$ f.t.

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5. Linear Equations [B]

a (4 marks)

M1 $Let the girl own g stickers; the boy owns 5g stickers$

M1 $g + 30 = 2(5g - 30)$
1M for forming the equation

M1 $g + 30 = 10g - 60 \Rightarrow 9g = 90 \Rightarrow g = 10$

A1 $Total = 10 + 50 = 60$ f.t.

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6. Percentages [B]

a (4 marks)

M1 $Let the cost be \backslash C; the marked price is \backslash(C + 90)$

M1 $(C + 90)(1 - 20$
1M for equating the selling price

M1 $0.8C + 72 = 1.2C \Rightarrow 0.4C = 72 \Rightarrow C = 180$

A1 $Marked price = \backslash 270$ f.t.
u-1 if the dollar sign is missing

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7. Polar Coordinates [B]

a (4 marks)

- A1 $\angle POQ = 125° - 65° = 60°$
- M1 $OP = OQ = r and \angle POQ = 60°, so \triangle OPQ is equilateral$
- A1 $OP = PQ = 27$ f.t.
- A1 $Perimeter = 27 \times 3 = 81$ f.t.

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8. Similar Triangles [B+]

a (2 marks)

- M1 $\angle CAE = \angle BDE (given); \angle AEC = \angle DEB (common angle)$
- PROOF $\therefore \triangle ACE \sim \triangle DBE (AAA)$

b (1 mark)

- A1 $AC^2 + AE^2 = 400 + 2304 = 2704 = 52^2 = CE^2, so YES, right-angled at A (converse of Pyth. thm)$

c (2 marks)

- M1 $Area of \triangle ACE = \frac{1}{2}(20)(48) = 480 \text{ cm}^2; ratio of sides = \frac{BD}{AC} = \frac{15}{20} = \frac{3}{4}$ f.t.
- A1 $Area of \triangle DBE = \left(\frac{3}{4}\right)^2 (480) = 270 \text{ cm}^2$ f.t.

  u-1 if unit missing

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9. Statistical Diagrams [B]

a (3 marks)

- M1 $Total number of students = 3 + k + 6 + 11 + 9 + 6 = 35 + k$
- M1 $\frac{3 + k + 6}{35 + k} = \frac{7}{20}$
- A1 $20k + 180 = 245 + 7k \Rightarrow 13k = 65 \Rightarrow k = 5$

b (2 marks)

- A1 $Range = 20 - 15 = 5; inter-quartile range = 19 - 17 = 2$ f.t.
- A1 $Standard deviation \approx 1.46$ f.t.

  r.t. 1.46

Section A(2) (35 marks)

10. Partial Variation [A-]

a (3 marks)

- M1 $Let f(x) = h + k(x+3)^2, where k \neq 0$
- M1 $h + 4k = -24 and h + 49k = 21 \Rightarrow 45k = 45$
- A1 $k = 1, h = -28 \Rightarrow f(0) = -28 + 9 = -19$

b (1 mark)

- A1 $y-intercept = f(0) + 3 = -19 + 3 = -16$ f.t.

c (2 marks)

- M1 $f(x) + 3 = (x+3)^2 - 25 = 0$ f.t.
- A1 $x + 3 = \pm 5 \Rightarrow x = 2 or x = -8$ f.t.

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11. Frequency Distribution [A-]

a (2 marks)

- M1 $Total number of children = 40; total tokens = 3 + 8 + 24 + 40 + 40 + 24 + 21 = 160$
- A1 $\text{Mean} = \frac{160}{40} = 4$

b (2 marks)

- M1 $Cumulative frequencies: 3, 7, 15, 25, 33, 37, 40; median = average of the 20th and 21st data = 4$
- A1 $Mode = 4 (highest frequency 10); hence they ARE equal$ f.t.

c (3 marks)

- A1 $(i) \frac{160 + 6n}{40 + n} = 5 \Rightarrow 160 + 6n = 200 + 5n \Rightarrow n = 40$ f.t.
- A1 $(ii) New median = 6; 33 data are at most 5, so the middle position must exceed 33: least n = 27$ f.t.
- A1 $(iii) Frequency of 6 becomes 4 + n, which must stay below 10: greatest n = 5$ f.t.

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12. Polynomials [A-]

a (3 marks)

M1 $p(x) = (x^2 + x + 1)(2x^2 - 13) + cx + c - 1$

M1 $p(3) = 0 \Rightarrow (9 + 3 + 1)(18 - 13) + 3c + c - 1 = 0$

A1 $65 + 4c - 1 = 0 \Rightarrow 4c = -64 \Rightarrow c = -16$

b (1 mark)

PROOF $p(-2) = (4 - 2 + 1)(8 - 13) + 16(2) - 17 = -15 + 15 = 0, sox + 2isafactor$ f.t.

c (3 marks)

M1 $p(x) = (x - 3)(x + 2)(2x^2 + 4x + 5)$ f.t.

M1 $For 2x^2 + 4x + 5 = 0 : \Delta = 4^2 - 4(2)(5) = 16 - 40 = -24 < 0$

A1 $Sothisquadratichastwonon - realroots; theclaimisNOTcorrect$ f.t.

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13. Circles and Locus [A-]

a (2 marks)

M1 $G = (8, 6)$

A1 $OG = \sqrt{8^2 + 6^2} = 10$

b (1 mark)

A1 $Radius = \sqrt{64 + 36 + 125} = 15 > 10 = OG, soYES, OliesinsideC$ f.t.

c (4 marks)

M1 $\Gamma istheperpendicularbisectorofOG : 4x + 3y - 25 = 0$

M1 $DistancefromG(8, 6)to\Gamma = \frac{|32 + 18 - 25|}{5} = 5$ f.t.

M1 $MN = 2\sqrt{15^2 - 5^2} = 2\sqrt{200} = 20\sqrt{2}$ f.t.

A1 $OG \perp MN, soarea = \frac{1}{2}(OG)(MN) = \frac{1}{2}(10)(20\sqrt{2}) = 100\sqrt{2}$ f.t.

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14. Mensuration [A]

a (2 marks)

M1 $\frac{1}{3}\pi r^2(24) = 800\pi \Rightarrow 8r^2 = 800$

A1 $r = 10$ cm

u-1 if unit cm missing

b (3 marks)

M1 $VolumeofX = \pi(10)^2(20) = 2000\pi; volumeofZ = 2000\pi + 800\pi = 2800\pi \text{ cm}^3$ f.t.

M1 $BaseradiusofZ = 20; \frac{1}{3}\pi(20)^2h = 2800\pi \Rightarrow h = 21 \text{ cm}$ f.t.

A1 $\frac{10}{20} = \frac{1}{2}but\frac{24}{21} = \frac{8}{7} \neq \frac{1}{2}, soYandZareNOTsimilar$ f.t.

c (3 marks)

M1 $CurvedsurfaceareaofX = 2\pi(10)(20) = 400\pi \text{ cm}^2$ f.t.

M1 $SlantheightofY = \sqrt{10^2 + 24^2} = 26; curvedsurfaceareaofY = \pi(10)(26) = 260\pi \text{ cm}^2$ f.t.

A1 $SlantheightofZ = \sqrt{20^2 + 21^2} = 29; curvedsurfaceareaofZ = \pi(20)(29) = 580\pi. Since 400\pi + 260\pi = 660\pi > 580\pi, theclaimisAGREED$ f.t.

Section B (35 marks)

15. Permutations and Probability [A-]

a (1 mark)

A1 $10! = 3628800$

b (3 marks)

M1 $Arrangethe6teachersfirst : 6!ways$

M1 $Choose4ofthe7gapsforthestudents : C_4^7 \times 4!ways$

A1 $P = \frac{6! \times C_4^7 \times 4!}{10!} = \frac{604800}{3628800} = \frac{1}{6}$ f.t.

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16. Linear Programming [A-]

a (3 marks)

M1 $SlopeofL_1 = \frac{12-8}{3-0} = \frac{4}{3}; L_1 : 4x - 3y + 24 = 0$

M1 $SlopeofL_2 = -\frac{3}{4}; L_2 : 3x + 4y - 57 = 0$ f.t.

A1 $4x - 3y + 24 \geq 0, 3x + 4y - 57 \leq 0, y \geq 0$ f.t.

b (2 marks)

M1 $VerticesofR : (-6, 0), (19, 0), (3, 12); values - 48, 152, -36$ f.t.

A1 $Leastvalue = -48$ f.t.

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17. Arithmetic Sequences and Logarithms [A]

a (2 marks)

M1 $7d = 58 - 23 = 35 \Rightarrow d = 5$

A1 $A(1) = 23 - 3(5) = 8$

b (5 marks)

M1 $\log_3[G(1)G(2)\cdots G(k)] = \log_3 G(1) + \log_3 G(2) + \cdots + \log_3 G(k)$
1M for using the law of logarithms

M1 $= A(1) + A(2) + \cdots + A(k) = \frac{k}{2}[2(8) + (k-1)(5)]$ f.t.

M1 $= \frac{k(5k+11)}{2} < 888 \Rightarrow 5k^2 + 11k - 1776 < 0$ f.t.

M1 $k < \frac{-11 + \sqrt{121 + 35520}}{10} \approx 17.78$ f.t.

A1 $Checking : sumfork = 17is816 < 888andfork = 18is909 > 888; greatestk = 17$ f.t.

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18. Trigonometry in 3-D [A]

a (2 marks)

M1 $TheperpendiculardistancebetweenADandBCisAB \sin \angle BAD = 48 \sin 55° \approx 39.32$

A1 $CD = \frac{39.32}{\sin 65°} \approx 43.4 \text{ cm}$
r.t. 43.4

b (2 marks)

M1 $AE = 48 \cos 55° \approx 27.53; takingEastheorigin, A = (0, 0, 27.53), C = (40, 39.32, 0), D = (58.33, 0, 0)$ f.t.

A1 $\cos \angle CAD = \frac{\vec{AC} \cdot \vec{AD}}{|\vec{AC}||\vec{AD}|} \Rightarrow \angle CAD \approx 39.9°$ f.t.
r.t. 39.9

c (3 marks)

M1 $AE \perp planeBCDE, sotherequiredangleis\angle AFEwhereFisthefootoftheperpendicularfromEtoCD$

M1 $Areaof\triangle ECD = \frac{1}{2}(39.32)(58.33) \Rightarrow EF = \frac{2 \times \text{area}}{CD} \approx 52.87$ f.t.

A1 $\tan \angle AFE = \frac{27.53}{52.87} \Rightarrow \angle AFE \approx 27.5° < 30°, soitdoesNOTexceed30°$ f.t.
r.t. 27.5

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19. Quadratic Functions and Inscribed Circles [A+]

a (2 marks)

M1 $f(x) = x^2 - (12k+12)x + (36k^2 + 72k + 41) = [x - (6k+6)]^2 - (6k+6)^2 + 36k^2 + 72k + 41$

A1 $f(x) = [x - (6k+6)]^2 + 5; Q = (6k+6, 5)$

b (1 mark)

A1 $12 - x = 6k + 6 \Rightarrow x = 6 - 6k; R = (6 - 6k, 5)$ f.t.

c (3 marks)

M1 $SlopeofQS = \frac{5 - (5 - 8k)}{(6k+6) - 6} = \frac{8k}{6k} = \frac{4}{3}$ f.t.

M1 $y - 5 = \frac{4}{3}[x - (6k+6)]$ f.t.

A1 $4x - 3y - 24k - 9 = 0$ f.t.

d (3 marks)

M1 $QR = 12kandtheheightfromSis8k, soQS = RS = \sqrt{(6k)^2 + (8k)^2} = 10k$ f.t.

M1 $Area = \frac{1}{2}(12k)(8k) = 48k^2; s = \frac{12k + 10k + 10k}{2} = 16k \Rightarrow r = \frac{48k^2}{16k} = 3k$ f.t.

A1 $ThecentreUliesonx = 6atdistance3kbelowQR : U = (6, 5 - 3k); C : (x - 6)^2 + (y - 5 + 3k)^2 = 9k^2$
f.t.

e (3 marks)

M1 $UT \perp QS, so\angle STU = 90°; STUVisarectangleifandonlyif\vec{ST} = \vec{VU}$ f.t.

M1 $TisthefootoftheperpendicularfromUtoQS : T = \left(6 + \frac{12k}{5}, 5 - \frac{24k}{5}\right); thenV = S + U - T = \left(6 - \frac{12k}{5}, 5 - \frac{31k}{5}\right)$
f.t.

A1 $6 - \frac{12k}{5} = -6givesk = 5, and5 - \frac{31(5)}{5} = -26; botharesatisfied, soYESitispossiblewhenk = 5$
f.t.